

Five-Stage Fourth-Order SDIRK

Skvortsov S54b non-geometric reference method

1 Butcher tableau

For $\dot{z} = f(t, z)$, the S54b tableau is

$$\begin{array}{c|ccccc} \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{8} & \frac{3}{8} & \frac{1}{4} & 0 & 0 \\ 1 & -\frac{3}{2} & \frac{3}{4} & \frac{2}{3} & \frac{1}{4} & 0 \\ 1 & 0 & \frac{1}{6} & \frac{2}{3} & -\frac{1}{12} & \frac{1}{4} \\ \hline & 0 & \frac{1}{6} & \frac{2}{3} & -\frac{1}{12} & \frac{1}{4} \end{array}.$$

All five diagonal entries equal $\gamma = 1/4$, so the method is singly diagonally implicit. The final row of A equals b^T , making the method stiffly accurate.

The stages are solved sequentially:

$$Z_i = z_n + h \sum_{j < i} a_{ij} f(t_n + c_j h, Z_j) + h \gamma f(t_n + c_i h, Z_i), \quad (1)$$

$$z_{n+1} = z_n + h \sum_{i=1}^5 b_i f(t_n + c_i h, Z_i). \quad (2)$$

Each stage therefore requires a nonlinear solve in the physical state dimension, rather than one simultaneous system containing all stages.

2 Fourth-order conditions

Writing $C = \text{diag}(c)$ and using componentwise powers, the coefficients satisfy

$$\begin{aligned} b^T \mathbf{1} &= 1, & b^T c &= \frac{1}{2}, & b^T c^2 &= \frac{1}{3}, & b^T A c &= \frac{1}{6}, \\ b^T c^3 &= \frac{1}{4}, & b^T C A c &= \frac{1}{8}, & b^T A c^2 &= \frac{1}{12}, & b^T A^2 c &= \frac{1}{24}. \end{aligned}$$

These are the Runge–Kutta conditions through order four. For sufficiently smooth problems the local truncation error is $\mathcal{O}(h^5)$ and the global error is $\mathcal{O}(h^4)$.

3 Deliberately non-geometric structure

A Runge–Kutta method is symplectic for general canonical Hamiltonian systems only if

$$b_i a_{ij} + b_j a_{ji} - b_i b_j = 0 \quad \text{for every } i, j.$$

The S54b coefficients do not satisfy this identity. For example, with $i = j = 5$ the left-hand side is $1/16$, not zero. Hence S54b is not a symplectic Runge–Kutta method.

Symmetry would require equality with the adjoint tableau, including

$$c_i = 1 - c_{6-i}, \quad b_i = b_{6-i}, \quad a_{ij} + a_{6-i, 6-j} = b_{6-j}.$$

Already $c_1 = 1/4 \neq 1 - c_5 = 0$, so S54b is not symmetric. These failures make the method a useful same-order control in long-time comparisons against geometric fourth-order methods.

4 Newton implementation

For a known right-hand side r_i , one stage solves

$$R_i(Z_i) = Z_i - r_i - h\gamma f(t_n + c_i h, Z_i) = 0.$$

Newton corrections satisfy

$$\left[I - h\gamma D_z f(t_n + c_i h, Z_i^{(k)}) \right] \delta_i^{(k)} = -R_i(Z_i^{(k)}).$$

For guiding-centre dynamics the implementation uses independent analytic 2×2 particle Jacobians. A centered finite-difference dense Jacobian is available for generic dynamical systems. Per-stage and per-step iterations, residual evaluations, residual norms, and effective tolerances are returned as diagnostics.

5 Reference

The coefficient table is the S54b method presented by L. M. Skvortsov, “Diagonally implicit Runge–Kutta methods for stiff problems,” *Computational Mathematics and Mathematical Physics*, 46(12), 2110–2123 (2006), DOI: 10.1134/S0965542506120098.

6 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \quad \kappa(t_0) = 0, \quad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance.

Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

```
src/simulation/formulations/state.py
```

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}) . Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.

The five converged physical stages supply the passive momentum quadrature with the tableau weights.